



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Infinite Series

Convergence of Series

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CONDUCTED BY

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Infinite Series

Infinite Series

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers. The infinite series generated by this sequence is the formal expression

$$\sum_{n=1}^{\infty} a_n.$$

Convergence of Infinite Series

Convergence of Infinite Series

We define the corresponding sequence of partial sums (s_m) by 

$$s_m = a_1 + a_2 + a_3 + \cdots + a_m,$$

and say that the series $\sum_{n=1}^{\infty} a_n$ converges to S if the sequence (s_m) converges to S . In this case, we write

$$\sum_{n=1}^{\infty} a_n = S.$$

If the sequence $\{S_N\}$ does not converge, the series is called divergent.

Problem

Discuss the convergence or otherwise of the series



$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)} + \cdots \text{ to } \infty.$$

Problem

Test the convergence or otherwise of the series

$$\sum_{n=0}^{\infty} (-1)^n.$$



Divergence

② Problem

Discuss the convergence or divergence of the harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}.$$

Convergence using MCT

② Problem

Test the convergence of the series:

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$



Algebraic Limit Theorem for Series

💡 Algebraic Limit Theorem for Series

Let $\{a_k\}$ and $\{b_k\}$ be sequences of real numbers such that

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k$$

are both convergent series. Then:

(i) The termwise sum series

$$\sum_{k=1}^{\infty} (a_k + b_k)$$

also converges, and its sum is the sum of the individual series:

$$\sum_{k=1}^{\infty} (a_k + b_k) = \sum_{k=1}^{\infty} a_k + \sum_{k=1}^{\infty} b_k.$$

(ii) For any constant $c \in \mathbb{R}$ (or $\mathbb{C}$), the scalar multiple series

$$\sum_{k=1}^{\infty} ca_k$$

also converges, and

$$\sum_{k=1}^{\infty} ca_k = c \sum_{k=1}^{\infty} a_k.$$

Cauchy Criterion for Series

💡 Cauchy Criterion for Series

The series

$$\sum_{k=1}^{\infty} a_k$$

converges if and only if, for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such

$$n > m \geq N, \quad \implies \quad |a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon.$$

Theorem

If the series

$$\sum_{k=1}^{\infty} a_k$$

converges, then the sequence (a_k) converges to 0; that is,

$$a_k \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$



Convergence of Geometric Series

Theorem

The geometric series

$$1 + x + x^2 + x^3 + \cdots \text{ to } \infty$$

satisfies the following:

- it converges if $-1 < x < 1$, i.e. if $|x| < 1$;
- it diverges if $x \geq 1$;
- it oscillates finitely if $x = -1$;
- it oscillates infinitely if $x < -1$.

Thank You!

We'd love your questions and feedback.

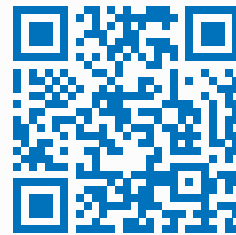
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(Lectures, walkthroughs, and course updates)



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References

- [1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.
- [2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.